Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/09/2023
PI Name:	Peters, Brian Ph.D.		
Project Title:	Orion Crew Performance and Safety Immediately Following Postflight Water Landings: Emergency Egress		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	KBR/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO 80JSC017N0001-Crew Health and Performance (FLAGSHIP1, OMNIBUS). Appendix A-Flagship1, Appendix B-Omnibus
Start Date:	07/27/2020	**End Date:**	09/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2022 per C. Ribeiro/JSC (Ed., 1/6/23)		
Key Personnel Changes/Previous PI:	Editor's Note: PI reports that Marissa Rosenberg, Ph.D. has taken another position outside of NASA, so is no longer a CoInvestigator directly associated with the study.		
COI Name (Institution):	Reschke, Millard Ph.D. (NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Caldwell, Erin M.S. (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation and augmentation of "Orion Crew Performance and Safety Immediately Following Postflight Water Landings: Emergency Egress (PI Reschke)" with new Principal Investigator Dr. Brian Peters. The decision to return to water landings in the Orion capsule presents significant safety issues that must be addressed. First, based on current models, it is anticipated that stable 2 capsule attitude upon splash down will occur more that 51% of the time and remain in that position for up to 20 min. Second, currently 100% of returning International Space Station (ISS) crewmembers (both Russian and US Orbital Segment (USOS)) have landing sickness, the severity of which is expected to increase due to wave motion coupled with exposure to a closed visual environment (cannot see a horizon). Third, strength, while compromised, is currently not believed to be an issue for emergency egress in long duration crewmembers (20+ days). However, crewmembers are vestibular and sensorimotor challenged to the point where coordinated walking, rising from a seat, or standing is difficult. Fourth, 100% of returning Soyuz crews when questioned about an unaided emergency egress from the Orion in the water have said that it will be catastrophic. The goal of the augmentation effort is to identify the optimal features of a device that visually presents gravitational reference cues to aid stability and re-adaptation that can be deployed in a capsule or aboard the recovery ship. The first specific aim is to enhance the existing capsule sensory-condition simulator (used in the existing Omnibus project) to include instrumented hand-holds, a compliant support surface and galvanic vestibular stimulation (GVS). Subjects in the previous investigation were destabilized by the wave motions but easily able to remain standing without support. They effectively used vestibular and proprioceptive cues to make the necessary adjustments to stay upright. The addition of the vestibular and proprioceptive challenges could easily increase the risk of falling. To mitigate this risk, hand holds will be added inside the simulator. The handles will also be instrumented with load cells to measure the amount of force imparted to them. Simple subject instructions, to minimize this force throughout each trial, will allow measures of the magnitude of the force to serve as the primary dependent variable when comparing test conditions. The second specific aim is to develop and test multiple presentation methods of an Earth-fixed visual reference. During this aim we will explore different visual aid options, both electrical and physical, to determine which can provide the best stabilizing effect. Special consideration is given to options that would be viable within the capsule, but it is important to remember that the stability challenges will also be present in an internal room on the recovery ship. Vertical and horizontal line projections will be compared as will the use of points of light instead of lines. Projected references could be problematic in the capsule so mechanical and custom-controlled light displays will also be explored. These conditions will be compared using a repeated measures design. The learning effects that complicated the interpretation of the previously collected data will be controlled by both a balanced test order and the inclusion of a control condition performed at the beginning and end of each test session.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Part I: This proposal effort sets out to develop a test that could be performed on returning crewmembers to better quantify astronauts' ability to egress the capsule after a water landing. A secondary goal is to compare the performance between a condition during which an Earth-fixed visual reference was projected onto the walls of the enclosure and another without the visual cue. Part II: The project was extended to advance the develop the Earth-fixed visual reference countermeasure hardware by determining the features of the visual that provide the better results. An Earth-fixed visual reference has applications for other motion platforms, including those on air, land, or sea.
Task Progress:	After delays associated with COVID restrictions, the Specific Aim 1 hardware modifications to the capsule sensory conditions simulator were completed. Specific Aim 2 was also completed with the data collection and analysis for 18 subjects using 5 different visual reference conditions; horizontal-only lines in the frontal and peripheral visual fields, horizontal + vertical lines in the frontal and peripheral fields and a star field pattern. Subjective verbal responses and quantitative data were collected. The preferred visual reference varied between subjects, but with the exception of one subject who appears to have misunderstood the task goal, all subjects reported that the presence of a visual reference helped them stabilize during the wave motion perturbations. A majority stated a preference for the presentations in the periphery, and the quantitative results confirmed that a single horizontal line presented in the periphery yielded the greatest stability benefit.
Bibliography Type:	Description: (Last Updated: 07/29/2021)